Business Analytics Case Study:Customer Behavior & Business Performance Analysis

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Target Audience: Executive Leadership Team, ABC Communications Ltd

1. Executive Summary

This consulting engagement presents an executive-level performance evaluation focused on customer retention metrics and operational revenue protection. Using a verified data pool consisting of **7,043 total customer records analyzed**, a diagnostic data analysis was conducted to evaluate health indicators across various product and billing segments.

The analysis uncovered a critically high baseline **churn rate of 26.54%**, representing an immediate loss of **1,869 customer portfolios**. This customer attrition rate poses a severe threat to long-term recurring revenue streams and increases customer acquisition costs. The report identifies key churn drivers, evaluates deep-dive transactional trends, and outlines immediate, data-backed strategic interventions to optimize customer lifetime value (LTV).

2. Deep-Dive Diagnostic Findings

A. Contractual Vulnerabilities & Milestone Risk

The structural design of account terms acts as a leading indicator of customer churn.

- **Month-to-Month Risk Exposure:** Customers on short-term Month-to-month billing cycles exhibit a catastrophic **42.71% churn rate**.
- **Long-Term Account Stability:** Accounts secured under One-year terms experience lower churn (**11.27%**), while Two-year contract holders drop to a structural baseline of just **2.83%**.
- **Lifecycle Tenure Lifespan:** A granular cohort review shows that the **0–12 Month band exhibits a 47.68% churn rate**. This risk footprint decreases significantly once a subscriber passes the 12-month milestone, dropping to **28.42%** in Year 2 and bottoming out at **8.20%** for mature accounts.

Interpretation: Short-term billing structures lack baseline switching costs. Retention messaging and client check-ins fail to capture users before they hit the high-risk, first-year drop-off threshold.

B. Premium Product Delivery Gaps & Payment Methods

An evaluation of internet infrastructure service types and billing mechanics reveals significant friction points within high-value segments.

- **Fiber Optic Attrition:** Premium Fiber Optic subscribers exhibit a highly concerning **41.89% churn footprint**, more than double the attrition seen on standard DSL setups (**18.96%**).
- **Transactional Friction:** Customers paying via **Electronic Check** experience a heavy **45.29% churn rate**, whereas automated transaction channels (Credit Cards and Bank Transfers) remain highly stable at ~15% churn.

Interpretation: Fiber Optic is a premium product. A ~42% loss rate indicates a clear gap between customer expectations and real-world execution. *Hypothesis to evaluate:* This trend may be driven by uncompetitive pricing packages, hardware onboarding friction, or technical stability issues relative to regional market competitors. Furthermore, manual payment models like Electronic Checks introduce monthly billing friction that prompts proactive customer exit.

3. Strategic Interventions & Recommendations

To plug the revenue leak and stabilize customer retention, management should deploy three targeted operational interventions:

- **Incentivize Contract Migration Programs:** Launch a proactive marketing campaign targeting Month-to-month accounts within their first 6 months. Offer structured promotional incentives (e.g., a 10% monthly service credit or complimentary streaming add-ons) to transition accounts into stable 1-Year or 2-Year contractual commitments.
- **Deploy a Fiber Optic Technical & Pricing Audit:** Task the engineering and customer support teams with testing the network stability of premium Fiber Optic pipelines. Run regional competitor price-matching audits to ensure product-market alignment and bundle free Technical Support to reduce technical onboarding hurdles.
- **Transition Users to Automated Auto-Pay:** Offer a one-time invoice credit (e.g., $5) to incentivize high-risk Electronic Check users to enroll in automated Credit Card or Bank Transfer billing pathways, eliminating routine transaction friction.

4. Supporting Visual Assets

Figure 1: Customer Loss Exposure by Account Term Structure

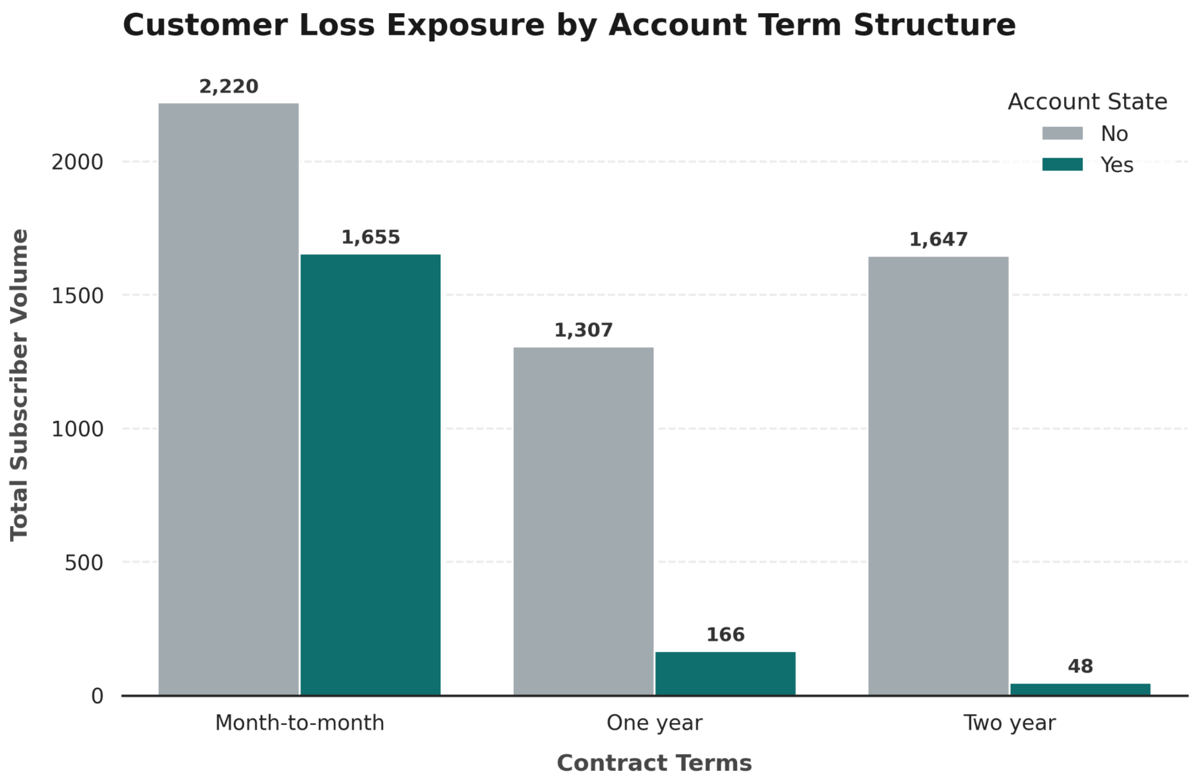


Figure 2: Strategic Performance Gaps Across Broadband Networks

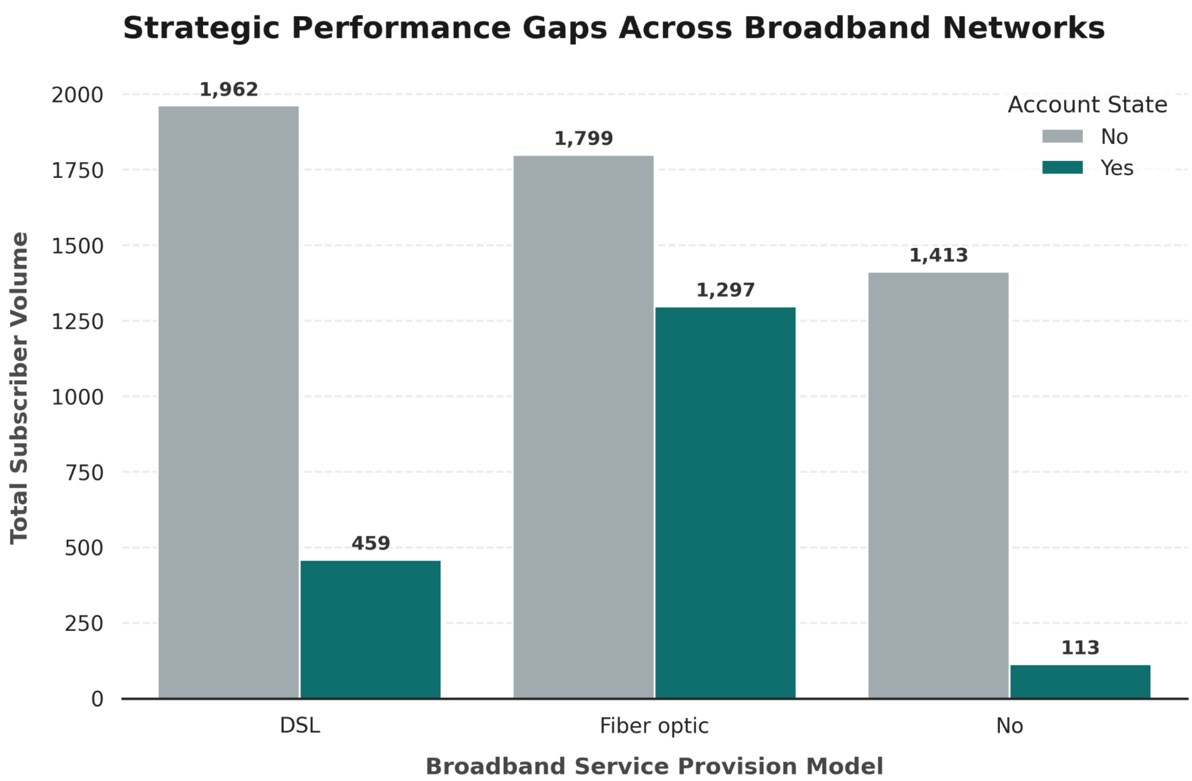
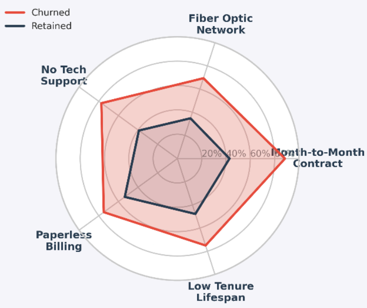


Figure 3: Multi-Dimensional Behavioral Matrix & Cohort Profiles

ABC Communications — Portfolio Retention Performance Brief

Behavioral Profile Grid



PORTFOLIO PERFORMANCE KPIS

7,043

Total Customer Accounts Analyzed

1,869

Total Accounts Churned

26.54%

Overall Customer Churn Rate

CHURN BY CONTRACT TYPE

Term Type	Retained	Churn Rate
Month-to-month	57.29%	42.71%
One year	88.73%	11.27%
Two year	97.17%	2.83%

CHURN BY INTERNET SERVICE

Service Model	Retained	Churn Rate
DSL	81.04%	18.96%
Fiber Optic	58.11%	41.89%
No Internet	92.60%	7.40%

CHURN BY PAYMENT METHOD

Payment Method	Retained	Churn Rate
Electronic Check	54.71%	45.29%
Mailed Check	80.80%	19.20%
Bank Transfer Auto	84.75%	15.25%
Credit Card Auto	83.27%	16.73%